

Coding & Solution

Date	7 July 2026
Team ID	XXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented solution. The Credit Card Approval Prediction System has been developed using a modular architecture with Flask for the web application and Machine Learning for prediction. The project follows clean coding practices, proper folder organization, reusable components, and documentation to ensure maintainability and scalability.

Instructions

- ✓ The source code is modular, organized, and follows Python coding standards.
- ✓ The application successfully implements the core objective of predicting credit card eligibility using Machine Learning.
- ✓ The project includes all required source code, trained model files, configuration files, datasets, and documentation.
- ✓ The complete project is maintained using Git and hosted on GitHub.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/NaveenEnikepalli/Credit_Card_Approval_Prediction
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost

Key Features Implemented	• Credit Card Eligibility Prediction • Data Preprocessing & Feature Engineering • Machine Learning Model Training & Comparison • XGBoost Prediction Engine • Flask Web Application • Responsive User Interface • Input Validation • Result Page with Eligibility Status, Confidence Score & Recommendations • Cloud Deployment using Render
Pending / Incomplete Features	• User Authentication (Future Enhancement) • Database Integration (Future Enhancement)
	• Admin Dashboard (Future Enhancement) • Applicant History Management (Future Enhancement)
Setup / Run Instructions	1. Clone the repository. 2. Create and activate a Python virtual environment. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Ensure the dataset and trained model files are available. 5. Run the application using <code>python app.py</code>. 6. Open <code>http://127.0.0.1:5000</code> in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

- The application follows a modular project structure with separate folders for templates, static assets, datasets, machine learning models, and utility modules.
- Multiple Machine Learning algorithms (Logistic Regression, Decision Tree, Random Forest, and XGBoost) were trained and evaluated to identify the best-performing model.
- XGBoost was selected as the final prediction model based on overall evaluation metrics.

- **The application provides real-time credit card eligibility prediction through a simple and responsive Flask web interface.**
- **The project is deployed on Render and maintained using GitHub for version control and collaboration.**
- **The architecture has been designed to support future enhancements such as authentication, database integration, and additional financial services.**